# When Do Attention-Head Ablations Support Causal Claims? Projection-Level Confounds, Floor Effects, and Matched Controls

**Juli Huang**
Stanford University
julih@stanford.edu

## Abstract

Attention-head ablation, replacing a head's output and measuring the resulting change in task performance, is a standard way to infer which components of a language model are causally responsible for a behavior. Using GPT-2 small as a case study, with replications on DistilGPT2 and Qwen2.5-1.5B-Instruct, we show that this recipe is fragile in ways that are rarely checked. First, a natural implementation of "zero out head $h$" (overwriting a channel slice of the attention block's *returned* output) does not ablate a single head, because the output projection has already mixed all heads. On a controlled induction-style copying task, this naive method is essentially uncorrelated with a corrected, pre-projection ablation (Pearson $r = 0.057$) and selects a disjoint top-5 set of heads. The confound recurs on both other models. Second, accuracy cannot register further degradation at a floor (0.0% on an off-target arithmetic task) and can miss confidence changes near a ceiling (99.5% on copying), while log-probability stays graded. Third, with a discovery/held-out split and 1,000 matched random-head and layer-matched control draws, the top-5 selected heads' held-out effect exceeds both controls (Monte Carlo $p = 0.001$), and it does so in all 20 random re-partitions. That effect, however, depends on the replacement value: under mean-ablation, the same GPT-2 heads' effect falls from 0.412 to 0.005 nats, and on indirect object identification zero-ablation shares only 2 of its top-5 heads with mean or resample ablation. Task specificity is not established on GPT-2: the contrast with arithmetic is inconclusive ($p = 0.078$; $p = 0.274$ at $N = 500$), and against a format-matched sequence-continuation control it disappears ($p = 0.567$). A single-head ablation result is not self-certifying. It requires verified projection semantics, a continuous metric away from floor and ceiling, held-out matched controls, and a stated replacement value checked against an on-distribution baseline.

## 1 Introduction

Interpretability researchers frequently ablate an individual attention head, replacing its contribution and re-measuring task performance, and interpret a resulting drop as evidence that the head is causally responsible for some computation. Whether this inference is valid depends on choices that are rarely audited: what, mechanically, gets replaced relative to the attention block's linear mixing structure; what it is replaced with; whether the evaluation metric has room to move at all; and whether the identified heads are validated against an appropriate null rather than just reported as the top of a ranking. This paper asks a narrow, mechanical question: **under what conditions does single-head ablation support a defensible causal interpretation**, in the sense of an effect that exceeds matched controls on held-out data?

40th Conference on Neural Information Processing Systems (NeurIPS 2026). Workshop: Interpretability as a Science.

We answer it by auditing and fixing a head-ablation implementation, designing a controlled induction-style copying task with behavioral headroom alongside an off-target floor task and a format-matched control task, and running a discovery/held-out pipeline with random and layer-matched controls. The primary model is GPT-2 small (124M parameters, 12 layers $\times$ 12 heads); DistilGPT2 and Qwen2.5-1.5B-Instruct serve as replications, and indirect object identification (IOI) on GPT-2 as an established circuit task. We report four findings (Section 4).

**(1, RQ0) The ablation implementation matters.** A hook registered on the whole attention module's output, a natural way to write "zero out head $h$", zeroes channels *after* the block's output projection has already mixed all heads together, so it does not isolate one head's contribution. On our controlled copying task, this naive method's per-head effect vector is essentially uncorrelated with a corrected, pre-projection version (Pearson $r = 0.057$), and the two methods' top-5 "important" heads share no head. The same holds on DistilGPT2 and Qwen2.5.

**(2, RQ1) Binary accuracy loses information at floor and near ceiling.** GPT-2 small scores 0.0% on an off-target arithmetic task and 99.5% on our on-target copying task. At the floor, accuracy cannot register further degradation. Near the ceiling, it can register a sufficiently large drop but may hide confidence changes that leave the predicted token unchanged. Teacher-forced log-probability of the gold continuation remains graded in both regimes.

**(3, RQ2) With proper controls, five heads show a held-out-replicated intervention effect that exceeds random and layer-matched baselines, but their task specificity is not established on GPT-2.** The per-head ranking is stable across splits (Spearman $\rho = 0.974$ over all 144 heads), and the top-5 discovery-selected heads' mean held-out effect clears both control distributions (Monte Carlo $p = 0.001$ each), in the original partition and in all 20 random re-partitions. Their contrast against an off-target arithmetic task is inconclusive ($p = 0.078$; $p = 0.274$ in a pre-committed $N = 500$ rerun), and against a format-matched sequence-continuation task it disappears ($p = 0.567$).

**(4) The replacement value changes which heads are selected.** Under mean-ablation, the five zero-selected GPT-2 heads have almost no held-out effect (0.005 vs. 0.412 nats), and reselection yields a disjoint set. On IOI, zero-ablation shares only 2 of its top-5 heads with mean or resample ablation. Baseline dependence is partial on DistilGPT2 and small on Qwen2.5, so it must be checked per model.

Our contribution is a worked, fully reproducible example (code, seeds, and every number traceable to a saved result file) showing that a positive *or* null single-head ablation result is not self-certifying: a causal interpretation additionally requires (i) verifying the intervention's projection semantics, (ii) a continuous metric that is not saturated, (iii) matched controls evaluated on data not used for head selection, and (iv) a stated replacement value checked against at least one on-distribution baseline.

## 2 Why naive head ablation can mislead

GPT-2's attention block computes a per-head output for each head, concatenates the per-head outputs, and applies a single dense output projection ($W_O$, `c_proj` in the standard `transformers` implementation) before adding the result to the residual stream. Elhage et al. [1] describe $W_O$ as half of each head's output-value (OV) circuit. Because $W_O$ spans the concatenated head dimension, each output channel can receive contributions from all heads. A hook that zeroes a contiguous, head-sized slice of the block's *returned* output (registered as a forward hook on the whole attention module) therefore does not remove one head's contribution. It removes an arbitrary linear combination of all heads' contributions to a subset of output channels, after mixing has already happened. The correct intervention instead zeroes the corresponding slice of $W_O$'s *input*, i.e., the concatenated, not-yet-mixed per-head values, which requires a pre-hook on the projection itself rather than a post-hook on the block.

We did not construct this as a strawman: it is the hook used by this project's own original head-ablation implementation, which we found when we audited it (preserved, unused for any reported result, as `src/legacy_invalid_ablation.py`). We use it as the "naive" method in our RQ0 comparison (Section 4). The methods use the same nominal head index, model, and input. Their activation interventions differ only in where they are applied within the block.

**Scope of the failure.** The failure concerns an implementation that treats a contiguous slice of the already-projected block output as if it were one head. The confound is a property of where the

intervention is placed, not of a particular library. TransformerLens [2] exposes a per-head, pre-projection hook (`hook_z`), which corresponds to our valid intervention, but it also exposes the block's mixed output (`hook_attn_out`), where slicing channels would reproduce the naive method. NNsight [3] exposes both the output of an attention module and the input of its output projection, so the same choice arises there. Our results do not show that these libraries implement the error, and we did not measure how often the naive placement occurs in published work. They show that the semantics of an intervention must be checked against the model's computational graph.

**Choice of replacement value.** Zero-ablation sets a head's output to a value it may never take during normal operation. Circuit analyses therefore often replace activations with dataset means [4], and Li and Janson [5] propose optimal ablation as an alternative. We use zero-ablation for the primary analysis because it is a common default, and we test whether the conclusions survive mean and resample replacement in Sections 4.4–4.5.

Separately, prior work has repeatedly cautioned that single-component interventions and attention weights are easy to over-interpret without controls: Jain and Wallace [6] show attention weights alone are often uncorrelated with feature importance, and Michel et al. [7] show many heads can be pruned post hoc without hurting overall performance, i.e., that a head being prunable does not mean it was doing nothing task-specific. More recent circuit-style work explicitly validates causal claims about specific components against matched baselines and held-out data [8, 4, 9]. Our contribution is narrower and more mechanical: we isolate the projection-level implementation confound specifically, and combine it with a floor/ceiling-aware metric and a minimal control battery in one end-to-end worked example.

## 3 Experimental design

**Models.** The primary model is GPT-2 small (124M parameters, 12 layers, 12 heads/layer, 144 heads total), the standard Hugging Face `transformers` implementation (v4.57.3, torch 2.8.0), run on CPU. Replications use DistilGPT2 (6 layers, 72 heads), which is distilled from GPT-2 and therefore not an independent architecture, and Qwen2.5-1.5B-Instruct [10] (28 layers, 12 query heads per layer, 336 heads), whose output projection is hooked in the same pre-projection position.

**Tasks.** *Controlled induction-style copying* (target task, expected headroom): synthetic four-word sequences of the form "$w_1\ w_2\ w_3\ w_4\ .\ w_1\ w_2\ w_3$" $\rightarrow$ "$\ w_4$", with $w_1, \ldots, w_4$ drawn without replacement from a pool of common English words verified to tokenize as exactly one byte-pair-encoding (BPE) token, a minimal instance of the induction-head phenomenon described by Olsson et al. [11] and traced to two-layer head composition by Elhage et al. [1]. We generated $N = 200$ examples (seed 0) and split them 50/50 into a *discovery* set (used only to select candidate heads) and a *held-out* set (used only to evaluate them), fixed before any head effect was computed. *Arithmetic* (off-target control task): "Q: $a$+$b$=" $\rightarrow$ "$\ a$+$b$", $a, b$ drawn uniformly from $[1, 20]$, $N = 200$ (seed 1), used only to test task-specificity of already-selected heads, never for selection. *Ordered-sequence continuation* (format-matched control task, added for the camera-ready version): "$x_1\ x_2\ x_3\ x_4\ .\ x_5\ x_6\ x_7$" $\rightarrow$ "$\ x_8$", where $x_1, \ldots, x_8$ are eight consecutive items of a familiar ordering (numerals 1–99, months, capital letters, number words one–twenty). Every non-wrapping window is used once, giving $N = 129$ examples (92 numerals, 19 letters, 13 number words, 5 months). The template, the period, and the single-token, space-prefixed answer match the copying task, but the answer never appears in the prompt, so it cannot be copied in context. It is predicted from knowledge of the ordering, a behavior attributed to "successor heads" [12]. Every generated example is checked so that the question's tokenization is an exact token-level prefix of question+answer (BPE merges are not always prefix-stable across concatenation). Examples failing this check are dropped (and, for the two sampled tasks, regenerated).

**Choice of tasks.** The tasks are deliberately simple. We use them to isolate methodological issues in head ablation, not to benchmark language understanding. Copying gives a predictable continuation with behavioral headroom. Arithmetic is an off-target diagnostic that also exposes accuracy at a floor, and it is not a matched control: it differs from copying in template, computation, answer tokenization, and baseline performance. The sequence task is a matched control in template, answer length, and baseline (Section 4.3). Simple tasks make the diagnostic results interpretable, but they limit conclusions about more complex behaviors, so we also test an established circuit task, indirect object identification (Section 4.5).

**Metrics.** *Teacher-forced exact match*: the greedy argmax token at every answer position must equal the gold token (binary). *Teacher-forced log-probability*: $\sum_t \log p(\text{gold token}_t \mid \text{prefix})$ over the answer's tokens (continuous, natural log). Both require a single forward pass per batch (no autoregressive generation).

**Ablation methods.** `naive_post_projection`: a forward hook on the attention module's output, zeroing a head-sized channel slice of the returned (already-projected) tensor. This is invalid and used only for the RQ0 comparison. `head_ablation`: a forward pre-hook on the output projection's input, zeroing the corresponding pre-mix slice. Every causal claim uses this pre-projection location. The primary analysis replaces the slice with zero. The robustness checks replace it with the slice's mean (*mean-ablation*; computed over all non-padding positions of the discovery split only and applied unchanged to every split, so held-out data never inform the intervention) or, for IOI only, with its activation on a corrupted prompt (*resample-ablation*; Section 4.5). These semantics are asserted by executable tests that run before the pipeline and ship with the code: on a small GPT-2 configuration they check that the pre-hook zeroes exactly the requested head's slice of the projection input and leaves every other channel unchanged, that removing the hook exactly restores the unablated output, and that the naive hook alters the logits while leaving that projection input untouched.

**Head selection.** For all 144 heads, we compute the mean $\Delta = \overline{\text{log-prob}}_{\text{baseline}} - \overline{\text{log-prob}}_{\text{ablated}}$ over a data split, using one batched forward pass per head (not per example). Candidate heads are the top-5 by $\Delta$ on the discovery split only.

**Controls.** *Random-head control*: 1,000 draws of 5 heads sampled uniformly without replacement from all 144. *Layer-matched control*: 1,000 draws that preserve the candidate set's size and layer composition while replacing every candidate identity, drawing each layer's replacements without replacement from that layer's non-candidate heads. Both exclusions matter whenever one layer supplies several candidates, as ours does: replacing each candidate independently would let a single head stand in for several candidates, and would let a candidate head enter its own control distribution. Both controls are built by resampling the already-computed 144-head $\Delta$ vector, so they cost no additional forward passes. These two controls ask whether the candidates' effect is unusual *among heads*. A separate *specificity* contrast asks whether it differs *between tasks*: the candidates' held-out copying $\Delta$ minus their $\Delta$ on a control task, compared with the same quantity for the 1,000 random head sets. Passing the head controls does not imply passing the task contrast.

**Statistics.** Where applicable, example-level point estimates are reported with percentile-bootstrap 95% confidence intervals (CIs; 2,000 resamples over examples). Control ranges are the 5th–95th percentiles of 1,000 draws. One-sided Monte Carlo $p$-values use the finite-sample correction $(k + 1)/(B + 1)$, where $k$ of $B = 1{,}000$ control draws match or exceed the candidate statistic. Rank stability across splits is measured with Spearman correlation over all 144 heads.

**Reproducibility.** Seeds (induction generation 0, arithmetic generation 1, control sampling 42), package versions, and the experiment configuration are recorded with every saved result. The full pipeline (`scripts/run_neurips_experiments.py`) reruns end-to-end in 198.99 s for the primary GPT-2 configuration on one 10-core Apple M5 CPU with 24 GB unified memory and no GPU (peak resident memory 1.94 GiB), 57.62 s for DistilGPT2, and 428.56 s for GPT-2 at $N = 500$ (3.22 GiB). Each of the three configurations was then rerun independently; all three reproduced their saved result files exactly, byte for byte, apart from the recorded timestamp. The camera-ready partition and sequence-control analyses (`scripts/run_camera_ready_robustness.py`) took 301.43 s (1.90 GiB) for GPT-2 and 82.08 s (1.63 GiB) for DistilGPT2. The software environment, model revisions, file hashes, storage, and verification commands are recorded in `neurips/experiment_results/reproducibility_provenance.md`. Every number in this paper is read from a result JSON file, and a script (`scripts/verify_paper_numbers.py`) checks the printed values against those files.

# 4 Results

## 4.1 RQ0: Does the ablation implementation matter?

On the controlled copying-task discovery split, the naive and correct per-head $\Delta$ vectors are nearly uncorrelated (Pearson $r = 0.057$, Spearman $\rho = 0.110$; scatter plot in Appendix A), with a mean absolute per-head disagreement of $0.067$ nats. The top-5 heads selected by each method share **zero**

heads. The same pattern holds on DistilGPT2 ($r = 0.001$) and on Qwen2.5-1.5B-Instruct ($r = 0.108$, $\rho = 0.176$), each with 0/5 top-head overlap (Appendices B and C). The methods use the same nominal head index, model, and input, but apply the activation intervention at different points in the block (Section 2). This is not a hypothetical failure mode: the naive intervention is a natural, easy-to-write hook, and it is the one this project originally used (Section 2), not one constructed to fail.

### 4.2 RQ1: Metric behavior at floor and near ceiling

GPT-2 small is at a behavioral floor on the arithmetic control task (0.0% teacher-forced exact match, 95% CI $[0.0\%, 0.0\%]$, $N = 200$) and near a behavioral ceiling on the controlled induction-style copying task (99.5%, CI $[98.5\%, 100.0\%]$). At the floor, binary accuracy cannot detect further degradation. Near the ceiling, it can detect a sufficiently large damaging effect, but a substantial confidence decrease may leave every argmax prediction unchanged. Log-probability remains graded: mean log-prob is $-7.44$ nats (CI $[-7.53, -7.35]$) on arithmetic and $-0.83$ nats (CI $[-0.90, -0.76]$) on the copying task (Table 1, Figure 1A), and, as shown next, individual-head ablation moves it substantially and reproducibly. This is why we select and validate candidate heads using log-probability rather than accuracy throughout. A post hoc check of one alternative answer format shows that the GPT-2 floor is not an artifact of the leading space in the gold answer: exact match remains 0.0% when the answer is written without it. This is not true for every model. Qwen2.5-1.5B-Instruct also scores 0.0%, because its tokenizer splits " 24" into a bare space token followed by digits and the model never predicts the bare space. Without the space it scores 23.0% (Appendix C). A floor in accuracy can therefore reflect the answer format rather than missing capability. Log-probability stays graded in either case (Qwen2.5: $-5.31$ nats).

### 4.3 RQ2: Specificity of selected-head intervention effects

The top-5 discovery-selected heads are Layer 0 Head 0, Layer 0 Head 7, Layer 1 Head 3, Layer 11 Head 0, and Layer 0 Head 10 (four in the first two layers, one in the last layer). Two distinct pieces of evidence support them. First, the per-head effect ranking is highly stable across the discovery and held-out splits (Spearman $\rho = 0.974$ over all 144 heads), so head ordering is not an artifact of the discovery sample. Second, the candidates' mean held-out effect, $0.412$ nats (CI $[0.353, 0.474]$), is far outside both the matched random-head control distribution (mean $0.029$, 5th–95th percentile range $[-0.033, 0.113]$; Monte Carlo $p = 0.001$, 0/1,000 draws matched or exceeded the candidates) and the layer-matched control distribution (mean $0.051$, 5th–95th percentile range $[0.001, 0.098]$; Monte Carlo $p = 0.001$; Figure 1B). This establishes a held-out-replicated intervention effect on the controlled copying task that exceeds what an arbitrary head, or an arbitrary head from the same layers, would produce. Whether that effect is specific to the copying task requires the separate contrast below.

Task specificity is not established (Figure 1C). The specificity statistic (held-out copying-task $\Delta$ minus arithmetic-control $\Delta$) is $0.139$ nats for the candidates, versus a random-control distribution with mean $0.011$ and 5th–95th percentile range $[-0.135, 0.165]$ (Monte Carlo $p = 0.078$). A single pre-committed rerun with $N = 500$ per task preserves the held-out effect over both head controls ($p = 0.001$ each) but weakens the contrast to $0.056$ nats ($p = 0.274$). We therefore treat the arithmetic contrast as inconclusive. Even a significant contrast against arithmetic would be hard to interpret, because the two tasks differ in template, computation, tokenization, and baseline performance.

**A format-matched control removes the GPT-2 contrast.** We therefore added the ordered-sequence task (Section 3), whose template, answer length, and baseline match copying (GPT-2: 96.9% exact match, $-0.57$ nats) but whose answer cannot be copied from the prompt. This analysis and the partition analysis below were specified in writing before they were run and are reported as run. Ablating the five GPT-2 candidates lowers the sequence-task log-probability by 0.447 nats, far more than random head sets do (mean 0.051, $p = 0.001$) and as much as it lowers held-out copying. Their specificity against this control is $-0.035$ nats (random-control mean $-0.023$, $p = 0.567$), and the candidates selected under mean-ablation behave the same way ($-0.138$ nats, $p = 0.882$). On GPT-2, the selected heads' effect is therefore not specific to in-context copying under either baseline. DistilGPT2 differs. Its candidates lower the sequence task by only 0.113 nats, no more than random head sets (mean 0.107, $p = 0.280$), so their specificity is 0.472 nats ($p = 0.001$; mean-ablation candidates 0.382, $p = 0.001$). For DistilGPT2, the effect is larger on in-context copying than on a

Table 1: Summary of all RQ1/RQ2 quantities. All values are computed from `gpt2_results.json` (Section 3). CIs are 95% percentile bootstrap over examples except where noted. "Control range" is the 5th–95th percentile of 1,000 control draws, not a CI.

| Quantity | Value | Interval | $N$ / draws |
|---|---|---|---|
| Copying-task accuracy (ceiling) | 0.995 | [0.985, 1.000] | 200 |
| Arithmetic accuracy (floor) | 0.000 | [0.000, 0.000] | 200 |
| Copying-task log-prob (nats) | $-0.83$ | $[-0.90, -0.76]$ | 200 |
| Arithmetic log-prob (nats) | $-7.44$ | $[-7.53, -7.35]$ | 200 |
| Candidate $\Delta$, held-out | 0.412 | [0.353, 0.474] | 100 |
| Random-control $\Delta$, held-out | 0.029 | $[-0.033, 0.113]^{\dagger}$ | 1000 |
| Layer-matched $\Delta$, held-out | 0.051 | $[0.001, 0.098]^{\dagger}$ | 1000 |
| Rank stability (disc. vs. held-out, $\rho$) | 0.974 | – | 144 heads |
| Candidate specificity | 0.139 | – | – |
| Random-control specificity | 0.011 | $[-0.135, 0.165]^{\dagger}$ | 1000 |

Monte Carlo $p$(candidate $\geq$ random control, held-out $\Delta$) $= 0.001$; $p$(candidate $\geq$ layer-matched, held-out $\Delta$) $= 0.001$; $p$(candidate $\geq$ random control, specificity) $= 0.078$. $^{\dagger}$5th–95th percentile.

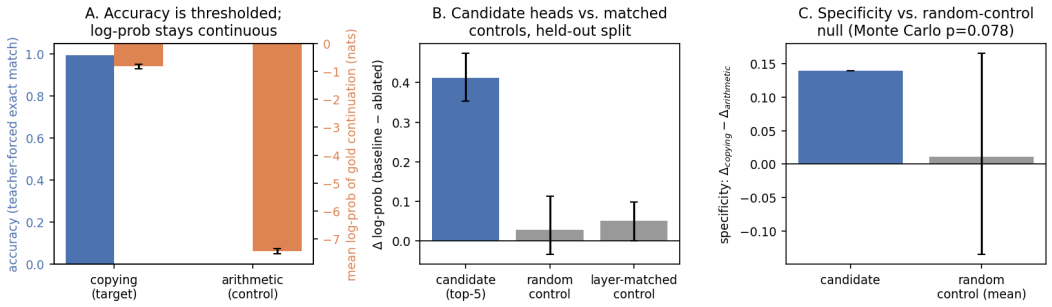



Figure 1: **A.** Accuracy cannot show further degradation at the arithmetic floor and may hide confidence changes near the copying-task ceiling. Log-probability stays continuous. **B.** The top-5 candidate heads' held-out effect far exceeds matched random-head and layer-matched control distributions (error bars: 95% CI for the candidate, 5th–95th percentile for controls). **C.** Specificity (copying minus arithmetic effect) is larger for candidates than the random-control mean, but the contrast is inconclusive (Monte Carlo $p = 0.078$; $p = 0.274$ at $N = 500$) and disappears against a format-matched control (Section 4.3).

format-matched task that cannot be solved by copying. This is evidence of a task-dependent effect, not of an induction mechanism, and its baseline match is looser (87.6% vs. 90.0% on copying).

**Sensitivity to the partition.** We repeated selection and evaluation over 20 random 100/100 partitions of the 200 copying examples (Appendix E). On GPT-2, the same five candidates are selected in 19 of 20 partitions and exceed both head controls in all 20 ($p = 0.001$), with a median held-out effect of 0.441 nats (range 0.368–0.469). The arithmetic specificity contrast, however, gives $p \leq 0.05$ in 9 of 20 partitions (median $p = 0.054$), so whether it crosses the conventional threshold depends on the partition. On DistilGPT2, the same five heads are selected in all 20 partitions, and both head controls and the arithmetic contrast give $p = 0.001$ in each. The partitions overlap, so this is a sensitivity analysis rather than 20 independent replications.

### 4.4 Robustness to the ablation baseline

Zero is an off-distribution value for a head's output, so an observed effect could reflect that distribution shift rather than the removal of a computation. In response to review, we repeated the full RQ2 pipeline with mean-ablation (Section 3), keeping the data, splits, selection rule, controls, seeds, and statistics unchanged (Table 2).

**The protocol still separates candidates from controls.** Under mean-ablation, the reselected candidates exceed both the random-head and the layer-matched controls on held-out data for both

Table 2: Mean-ablation robustness check ($N = 200$ per task; $\Delta$ in nats). The first block compares per-head effects under the two baselines. The second repeats RQ2 with candidates reselected under mean-ablation. The last block evaluates the zero-selected candidates under both baselines and was computed post hoc from the saved per-head effects. Intervals are 95% percentile-bootstrap CIs over examples; $^{\dagger}$5th–95th percentile of 1,000 control draws.

| | GPT-2 small | | DistilGPT2 | |
|---|---|---|---|---|
| Quantity | Value | Interval | Value | Interval |
| Zero vs. mean per-head $\Delta$: Pearson $r$ | 0.424 | – | 0.553 | – |
| Zero vs. mean per-head $\Delta$: Spearman $\rho$ | 0.495 | – | 0.871 | – |
| Top-5 overlap, mean- vs. zero-selected | 0/5 | – | 1/5 | – |
| Candidate $\Delta$, held-out | 0.176 | [0.157, 0.194] | 0.445 | [0.388, 0.508] |
| Random-control $\Delta$, held-out | 0.016 | $[-0.034, 0.057]^{\dagger}$ | 0.060 | $[-0.012, 0.172]^{\dagger}$ |
| Layer-matched $\Delta$, held-out | 0.009 | $[-0.071, 0.058]^{\dagger}$ | 0.030 | $[-0.024, 0.067]^{\dagger}$ |
| Rank stability $\rho$ | 0.948 | – | 0.960 | – |
| $p$: random / layer-matched / specificity | 0.001 / 0.001 / 0.014 | | 0.001 / 0.001 / 0.001 | |
| Zero-selected heads, held-out $\Delta$ under zero | 0.412 | – | 0.586 | – |
| Zero-selected heads, held-out $\Delta$ under mean | 0.005 | – | 0.182 | – |

models (Monte Carlo $p = 0.001$ for each comparison), and the per-head ranking remains stable across splits (Spearman $\rho = 0.948$ for GPT-2, $0.960$ for DistilGPT2).

**The selected heads and their effect depend on the baseline.** On GPT-2, per-head effects under the two baselines are only moderately correlated on the discovery split (Pearson $r = 0.424$, Spearman $\rho = 0.495$). The mean-ablation candidates (Layer 10 Heads 11 and 0, Layer 7 Head 6, Layer 0 Heads 9 and 3) share no head with the zero-ablation candidates, and their held-out effect is smaller (0.176 nats, CI $[0.157, 0.194]$). In a post hoc comparison computed from the saved per-head vectors, the five zero-selected GPT-2 heads have a mean held-out effect of only 0.005 nats under mean-ablation, compared with 0.412 nats under zero-ablation, and below the mean of the mean-ablation random control (0.016). The RQ2 effect of these particular heads is therefore consistent with replacing their output by an off-distribution value rather than with removing information the copying task needs. On DistilGPT2 the baselines agree more closely ($r = 0.553$, $\rho = 0.871$, 1/5 overlap), and the zero-selected heads keep part of their effect (0.586 to 0.182 nats). On Qwen2.5-1.5B-Instruct the same five heads are selected under both baselines (5/5 overlap) and keep most of their effect (0.208 to 0.155 nats; Appendix C). Under mean-ablation the GPT-2 specificity contrast against arithmetic gives $p = 0.014$ (0.168 nats versus a random-control mean of 0.021), but it concerns a different head set, comes from a single run, and inherits the arithmetic mismatch, so we do not treat it as evidence of task specificity.

### 4.5 An established circuit task: indirect object identification

To test whether baseline dependence also arises on a natural-language circuit task, we ran indirect object identification (IOI; 4) on GPT-2 small: 200 prompts from the 15 templates of Wang et al. [4] with single-token names, places, and objects, a 100/100 discovery/held-out split, and the logit difference between the indirect object and the subject as the metric. GPT-2 prefers the indirect object on 97% (discovery) and 100% (held-out) of prompts. We compared three replacement values at the same pre-projection location: zero, the discovery-split mean, and a *resample* value, the head's activation on the "ABC" variant (the second subject replaced by a third name) of a different prompt with the same template. Selection, controls, and statistics are those of RQ2. Under every baseline the selected heads exceed both matched controls ($p = 0.001$ each). The baselines disagree on which heads matter: zero shares only 2 of its top-5 heads with mean or with resample, whereas mean and resample share 4 of 5 (Spearman $\rho$ over all 144 heads: zero vs. mean $0.74$, zero vs. resample $0.47$, mean vs. resample $0.65$). Zero-ablation selects three layer-0/1 heads that neither on-distribution baseline selects, and its held-out effect is 68% larger (1.55 vs. 0.92 and 0.93). Under every baseline the top-5 includes the S-inhibition heads 8.6 and 8.10 reported by Wang et al. [4], and their head 7.9 is in the top-5 under mean and resample (Appendix D).

## 5 Discussion and limitations

**Scope of the causal claim.** We do not claim these five heads "compute induction" in a mechanistic sense. Using the causal-claim ladder (correlation $<$ decodability $<$ intervention effect $<$ specific causal contribution $<$ mechanistic role), our evidence supports a held-out-replicated intervention effect that exceeds matched controls (the third rung), but only relative to a stated replacement value. The zero-selected GPT-2 heads reach this rung under zero-ablation and not under mean-ablation. The Qwen2.5 heads reach it under both. No GPT-2 specificity contrast supports the fourth rung. A mechanistic account would require further evidence, such as activation patching, input-dependence analysis, or directional interventions.

**Task-control mismatch.** The arithmetic control differs from copying in template, computational demands, tokenization, and baseline performance, and it sits at an accuracy floor. A difference in ablation effects between the two tasks therefore cannot be attributed uniquely to a copying-specific computation, and we treat the arithmetic contrast as a limited diagnostic. The sequence-continuation control (Section 4.3) matches the template, answer length, and baseline. It removes the GPT-2 contrast but not the DistilGPT2 one, and it is still only one alternative rule.

**Ablation baseline.** We tested zero, mean, and (for IOI) resample replacement. We did not test noise injection or optimal ablation [5]. Because the baseline changed the selected heads on GPT-2 and IOI but barely on Qwen2.5, whether it matters has to be checked for each model and task.

**Model scope.** DistilGPT2 is distilled from GPT-2, so it is not an independent replication. On it, the naive and correct effects are again near-uncorrelated ($r = 0.001$, 0/5 overlap), the arithmetic floor recurs, and the selected heads exceed both matched controls (Appendix B). Copying accuracy is lower (90.0%), so the near-ceiling example is specific to GPT-2. Qwen2.5-1.5B-Instruct is a different architecture family and reproduces the projection confound and the matched-control separation (Appendix C). We did not test models larger than 1.5B parameters, and the new partition and sequence-control analyses were run on GPT-2 and DistilGPT2 only, because one Qwen pipeline run took 3.3 CPU-hours.

**Partitions and sample size.** The partition analysis reuses the same 200 copying examples and uses zero-ablation only. The sequence control has 129 examples, 92 of them numerals.

**Broader implication.** Each check changed a conclusion: the naive hook selected disjoint heads; accuracy hid information at the floor and near the ceiling; the specificity contrast blocked a task-specificity claim that the larger GPT-2 run does not support; and the replacement value changed which heads were selected and, for GPT-2, removed the selected heads' effect. A single-head ablation report should therefore state the intervention point with a test that it isolates one head, use a continuous metric beside accuracy, select on held-out-separated data against matched controls, and report the replacement value together with at least one on-distribution baseline.

## 6 Conclusion

A head ablation is not self-certifying evidence of a specific causal role. Hook placement can ablate the wrong quantity, accuracy is censored at a floor and coarse near a ceiling, and held-out matched controls separate a replicating effect from arbitrary disruption. That effect can still depend on the replacement value, and task specificity can still fail to replicate.

## References


[1] Nelson Elhage, Neel Nanda, Catherine Olsson, Tom Henighan, Nicholas Joseph, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Nova DasSarma, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. A mathematical framework for transformer circuits. Transformer Circuits Thread, 2021. URL `https://transformer-circuits.pub/2021/framework/index.html`.

[2] Neel Nanda and Joseph Bloom. Transformerlens. `https://github.com/TransformerLensOrg/TransformerLens`, 2022.

[3] Jaden Fiotto-Kaufman, Alexander R. Loftus, Eric Todd, Jannik Brinkmann, Koyena Pal, Dmitrii Troitskii, Michael Ripa, Adam Belfki, Can Rager, Caden Juang, Aaron Mueller, Samuel Marks, Arnab Sen Sharma,

Francesca Lucchetti, Nikhil Prakash, Carla Brodley, Arjun Guha, Jonathan Bell, Byron C. Wallace, and David Bau. NNsight and NDIF: Democratizing access to open-weight foundation model internals. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2407.14561.

[4] Kevin Ro Wang, Alexandre Variengien, Arthur Conmy, Buck Shlegeris, and Jacob Steinhardt. Interpretability in the wild: A circuit for indirect object identification in GPT-2 small. In *International Conference on Learning Representations (ICLR 2023)*, 2023.

[5] Maximilian Li and Lucas Janson. Optimal ablation for interpretability. In *Advances in Neural Information Processing Systems 37*, pages 109233–109282, 2024. URL https://arxiv.org/abs/2409.09951.

[6] Sarthak Jain and Byron C. Wallace. Attention is not explanation. In *Proceedings of NAACL-HLT 2019*, pages 3543–3556, 2019.

[7] Paul Michel, Omer Levy, and Graham Neubig. Are sixteen heads really better than one? In *Advances in Neural Information Processing Systems 32 (NeurIPS 2019)*, pages 14014–14024, 2019.

[8] Kevin Meng, David Bau, Alex Andonian, and Yonatan Belinkov. Locating and editing factual associations in GPT. In *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022.

[9] Arthur Conmy, Augustine N. Mavor-Parker, Aengus Lynch, Stefan Heimersheim, and Adrià Garriga-Alonso. Towards automated circuit discovery for mechanistic interpretability. In *Advances in Neural Information Processing Systems 36 (NeurIPS 2023)*, pages 16318–16352, 2023.

[10] Qwen Team. Qwen2.5 technical report. *Computing Research Repository*, arXiv:2412.15115, 2024. URL https://arxiv.org/abs/2412.15115.

[11] Catherine Olsson, Nelson Elhage, Neel Nanda, Nicholas Joseph, Nova DasSarma, Tom Henighan, Ben Mann, Amanda Askell, Yuntao Bai, Anna Chen, Tom Conerly, Dawn Drain, Deep Ganguli, Zac Hatfield-Dodds, Danny Hernandez, Scott Johnston, Andy Jones, Jackson Kernion, Liane Lovitt, Kamal Ndousse, Dario Amodei, Tom Brown, Jack Clark, Jared Kaplan, Sam McCandlish, and Chris Olah. In-context learning and induction heads. *arXiv preprint arXiv:2209.11895*, 2022.

[12] Rhys Gould, Euan Ong, George Ogden, and Arthur Conmy. Successor heads: Recurring, interpretable attention heads in the wild. In *The Twelfth International Conference on Learning Representations (ICLR 2024)*, 2024. URL https://openreview.net/forum?id=kvcbV8KQsi.

## A Supplementary figure: RQ0 per-head comparison

Figure 2 plots the per-head effect (correct pre-projection ablation vs. naive post-projection ablation) for all 144 heads on the controlled copying-task discovery split, corresponding to the summary statistics in Section 4.1.

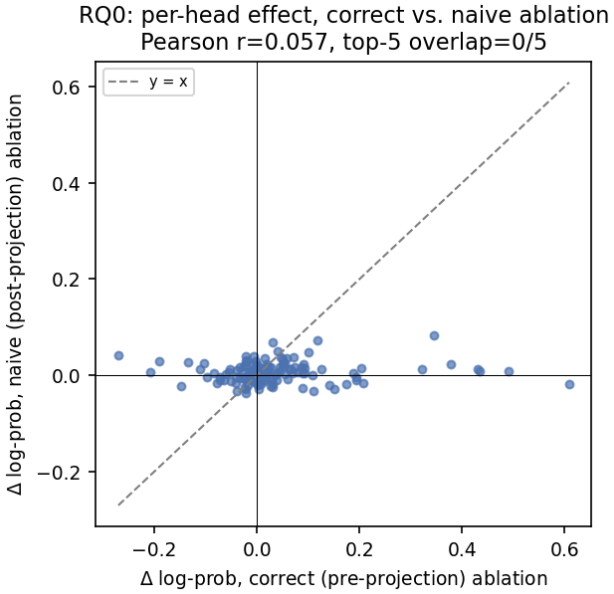



Figure 2: Per-head $\Delta$ log-prob under the correct (pre-projection) vs. naive (post-projection) ablation method, all 144 heads, controlled copying-task discovery split. A valid replication of the naive method would lie near the $y = x$ line. Instead the two methods are nearly uncorrelated (Pearson $r = 0.057$).

## B Replication on a second model (DistilGPT2)

As a first robustness check (Section 5), we reran the identical pipeline (Section 3), unchanged except for the model, on DistilGPT2 (6 layers, 12 heads/layer, 72 heads total). All data, metrics, ablation methods, splits, and control procedures are exactly as in Section 3, and only the model differs. Results are in Table 3.

The candidate heads found on DistilGPT2 (Layer 0 Heads 1, 7, 9; Layer 3 Head 2; Layer 5 Head 0) are, as expected, different indices from GPT-2 small's. The two models are different parameter sets, so this is not evidence the same head performs the same role across models, only that the same *protocol* (discovery/held-out selection against matched controls) again separates a real, held-out-replicated effect from the control distributions. We report this as a second data point for the paper's methodological claim, not as evidence about a shared or universal "induction circuit" across GPT-2 and DistilGPT2. DistilGPT2 is distilled from GPT-2, so it is not an independent model. Its arithmetic specificity contrast ($p = 0.001$) comes from a single run with no larger rerun and is subject to the same task mismatch as GPT-2's (Section 5), so we do not treat it as evidence of task specificity.

## C Replication on a different architecture (Qwen2.5-1.5B-Instruct)

We reran the identical pipeline, with both zero- and mean-ablation and unchanged except for the model, on Qwen2.5-1.5B-Instruct [10]. Its pre-projection hook sits on the input of `o_proj`. Executable tests on a small random Qwen2 configuration assert the same intervention semantics as for GPT-2. The run took 11,976 CPU-seconds with 9.0 GiB peak resident memory and was performed once (Table 4). The projection confound recurs ($r = 0.108$, 0/5 overlap). The zero-ablation candidates (Layer 8 Head 3, Layer 14 Head 3, Layer 18 Head 8, Layer 13 Head 4, Layer 2 Head 3) exceed both

Table 3: DistilGPT2 replication of Table 1, with the same procedure and $N$s. The intervention-semantic and matched-control findings replicate, and the arithmetic floor recurs. Copying accuracy is lower (90.0%), so GPT-2's near-ceiling example does not replicate. The specificity comparison gives Monte Carlo $p = 0.001$.

| Quantity | Value | Interval | $N$ / draws |
|---|---|---|---|
| Copying-task accuracy | 0.900 | [0.855, 0.940] | 200 |
| Arithmetic accuracy (floor) | 0.000 | [0.000, 0.000] | 200 |
| Copying-task log-prob (nats) | $-0.96$ | $[-1.07, -0.85]$ | 200 |
| Arithmetic log-prob (nats) | $-7.38$ | $[-7.50, -7.25]$ | 200 |
| RQ0: Pearson $r$(naive, correct) | 0.001 | – | 72 heads |
| RQ0: top-5 head overlap | 0/5 | – | – |
| Candidate $\Delta$, held-out | 0.586 | [0.511, 0.661] | 100 |
| Random-control $\Delta$, held-out | 0.080 | $[-0.026, 0.231]^{\dagger}$ | 1000 |
| Layer-matched $\Delta$, held-out | 0.054 | $[-0.017, 0.128]^{\dagger}$ | 1000 |
| Rank stability ($\rho$, disc. vs. held-out) | 0.957 | – | 72 heads |
| Candidate specificity | 0.492 | – | – |
| Random-control specificity | 0.023 | $[-0.159, 0.222]^{\dagger}$ | 1000 |

Monte Carlo $p$(candidate $\geq$ random control, held-out $\Delta$) $= 0.001$; $p$(candidate $\geq$ layer-matched, held-out $\Delta$) $= 0.001$; $p$(candidate $\geq$ random control, specificity) $= 0.001$. $^{\dagger}$5th–95th percentile.

controls, and under mean-ablation the same five heads are selected and still exceed both controls. The specificity contrast against arithmetic is significant under both baselines ($p = 0.004$ and $0.007$), subject to the task-mismatch caveat. Exact match on arithmetic is 0.0%, but 23.0% when the answer is written without its leading space (Section 4.2).

Table 4: Qwen2.5-1.5B-Instruct ($N = 200$ per task, 336 heads). Intervals are 95% percentile-bootstrap CIs over examples; $^{\dagger}$5th–95th percentile of 1,000 control draws.

| | Zero-ablation | | Mean-ablation | |
|---|---|---|---|---|
| Quantity | Value | Interval | Value | Interval |
| Copying-task accuracy | 0.990 | [0.975, 1.000] | | |
| Arithmetic accuracy | 0.000 | [0.000, 0.000] | | |
| Copying-task log-prob (nats) | $-0.38$ | $[-0.43, -0.34]$ | | |
| Arithmetic log-prob (nats) | $-5.31$ | $[-5.44, -5.17]$ | | |
| RQ0: $r$ / $\rho$ (naive vs. correct) | 0.108 / 0.176 | – | | |
| RQ0: top-5 head overlap | 0/5 | – | | |
| Zero vs. mean $\Delta$: $r$ / $\rho$ | | | 0.925 / 0.756 | – |
| Top-5 overlap with zero-selected | | | 5/5 | – |
| Candidate $\Delta$, held-out | 0.208 | [0.176, 0.243] | 0.155 | [0.129, 0.182] |
| Random-control $\Delta$, held-out | 0.012 | $[-0.010, 0.046]^{\dagger}$ | 0.006 | $[-0.007, 0.031]^{\dagger}$ |
| Layer-matched $\Delta$, held-out | 0.004 | $[-0.012, 0.019]^{\dagger}$ | 0.003 | $[-0.008, 0.015]^{\dagger}$ |
| Rank stability $\rho$ | 0.927 | – | 0.850 | – |
| Candidate specificity | 0.376 | – | 0.349 | – |
| Random-control specificity | $-0.007$ | $[-0.223, 0.205]^{\dagger}$ | 0.001 | $[-0.203, 0.189]^{\dagger}$ |
| $p$: random / layer / specificity | 0.001 / 0.001 / 0.004 | | 0.001 / 0.001 / 0.007 | |

## D Indirect object identification details

Table 5 gives the selected heads and the ranks of the heads that Wang et al. [4] identify, under each replacement value (Section 4.5). The resample source for each prompt is drawn uniformly (fixed seed) from the other prompts of the same template within the same split, so source and target have identical token length and every position plays the same syntactic role. Name mover 9.6 ranks 141st or lower under all three baselines, so ablating it alone increases the logit difference. Wang et al. [4] report backup heads that compensate when name movers are removed. Our single-head result is consistent with such compensation, although we did not test it. The run is saved in `results/naacl_ioi/gpt2_ioi_ablation_methods.json`.

Table 5: IOI on GPT-2 small (200 prompts, 100/100 discovery/held-out). Head $\ell.h$ is layer $\ell$, head $h$; top-5 heads in rank order. Held-out effect: mean drop in logit difference (IO minus subject) for the five selected heads, with 95% bootstrap CI. Ranks are by mean effect over all 200 prompts; rank 144 is the most negative effect.

| | Zero | Mean | Resample |
|---|---|---|---|
| Top-5 heads | 0.10, 1.3, 8.6, 8.10, 0.7 | 8.6, 5.5, 8.10, 7.9, 5.9 | 8.6, 8.10, 5.5, 7.9, 7.3 |
| Held-out effect | 1.55 [1.42, 1.69] | 0.92 [0.84, 1.01] | 0.93 [0.84, 1.02] |
| $p$ vs. random / layer-matched | .001 / .001 | .001 / .001 | .001 / .001 |
| Rank stability $\rho$ | 0.94 | 0.97 | 0.91 |
| Rank (of 144) of heads reported by Wang et al. [4] | | | |
| S-inhibition 7.3 / 7.9 / 8.6 / 8.10 | 14/6/3/5 | 7/4/1/2 | 5/4/1/2 |
| Name mover 9.6 / 9.9 / 10.0 | 141/26/15 | 141/16/9 | 142/12/9 |
| Negative name mover 10.7 / 11.10 | 144/143 | 144/143 | 144/143 |

## E Partition sensitivity

Table 6 summarizes the 20 random 100/100 partitions of the 200 copying examples (permutation seeds 1000–1019). For each partition we reselect the top-5 heads on its discovery half and repeat the RQ2 statistics on its held-out half, using the zero-ablation per-head effects computed once on all 200 examples. Restricted to the original partition, these effects equal the saved per-head values exactly. Results are in `<model>_camera_ready_robustness.json`.

Table 6: Partition sensitivity over 20 random 100/100 discovery/held-out partitions (zero-ablation). Ranges are minimum–maximum over partitions. Overlap is with the candidates of the original partition.

| | GPT-2 small | DistilGPT2 |
|---|---|---|
| Partitions with the original top-5 (5/5 overlap) | 19/20 | 20/20 |
| Candidate held-out $\Delta$, median [range] | 0.441 [0.368, 0.469] | 0.578 [0.535, 0.643] |
| Rank stability $\rho$, median [range] | 0.966 [0.937, 0.979] | 0.961 [0.923, 0.981] |
| Partitions with $p \leq 0.05$ vs. random / layer-matched | 20/20 / 20/20 | 20/20 / 20/20 |
| Arithmetic specificity, median [range] | 0.168 [$-0.014$, 0.196] | 0.484 [0.441, 0.549] |
| Partitions with arithmetic specificity $p \leq 0.05$ | 9/20 | 20/20 |
| Arithmetic specificity $p$, median [range] | 0.054 [0.039, 0.650] | 0.001 [0.001, 0.001] |

## F Reproducibility details

All results are produced by `scripts/run_neurips_experiments.py` and saved to `neurips/experiment_results/gpt2_results.json` (main results, Table 1) and `distilgpt2_results.json` (replication, Table 3). These files record the device, torch (2.8.0) and transformers (4.57.3) versions, all random seeds, per-split example counts, the number of bootstrap resamples (2,000) and control draws (1,000), the full per-head $\Delta$ vector for every split and method, and per-example log-probabilities for every baseline. Exact model revisions and hashes are recorded in `neurips/experiment_results/reproducibility_provenance.md`. A separate one-time, higher-power rerun of the main GPT-2 pipeline with $N{=}500$ (rather than 200) per task, run once and reported regardless of outcome, is saved to `gpt2_n500_results.json`. Its preserved held-out effect and weaker specificity contrast are reported in Section 4.3. Figures are produced by `scripts/make_figures.py`, which reads only the primary JSON file and performs no additional computation. A claim-to-evidence map (`neurips/claim_evidence_map.md`) lists, for every quantitative statement in this paper, the exact JSON field it was read from. Every file named in this appendix, together with the pipeline source, the figure script, and a README giving the pinned environment and the exact command for each reported run, is available at `https://github.com/TheClassicTechno/InterpretabilityNLPProject`. No dataset is shipped: all task families are generated in memory by `src/dataset.py` from the seeds recorded in each result file (the sequence-continuation task is deterministic).

The analyses added for the camera-ready version are saved separately and leave the original result files unchanged: `{gpt2,distilgpt2}_meanablation_results.json` and `mean_vs_zero_posthoc.json` (Section 4.4), `qwen2.5-1.5b-instruct_meanablation_results.json` and `arithmetic_format_diagnostic.json` (Appendix C), `results/naacl_ioi/` (Section 4.5), and `{gpt2,distilgpt2}_camera_ready_robustness.json` (Section 4.3, Appendix E). The design of the partition and sequence-control analyses was fixed in `neurips/camera_ready_design.md` before they were run.

## NeurIPS Paper Checklist

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: [Yes]

   Justification: The abstract and introduction report the four findings in Section 4: a projection-level implementation confound, asymmetric limits of binary accuracy at a floor and near a ceiling, held-out effects exceeding matched controls, and the dependence of selected heads and their effects on the replacement value. They state that GPT-2 task specificity was not established: the arithmetic contrast is inconclusive ($p = 0.078$; $p = 0.274$ at $N = 500$) and the contrast against a format-matched control is absent ($p = 0.567$). The claims cover GPT-2 small, DistilGPT2, and Qwen2.5-1.5B-Instruct on synthetic tasks, plus IOI on GPT-2.

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: [Yes]

   Justification: Section 5 discusses the causal claim's scope and its dependence on the replacement value, the mismatch between the arithmetic control and copying, the ablation baselines not tested (noise, optimal ablation), the model scope (DistilGPT2 is not independent of GPT-2; nothing above 1.5B parameters; partition and sequence-control analyses not run on Qwen2.5), and the reuse of one set of 200 copying examples and a numeral-dominated sequence control.

   Guidelines:

   - The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
   - The authors are encouraged to create a separate "Limitations" section in their paper.
   - The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
   - The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
   - The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
   - The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
   - If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.

- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: [N/A]

   Justification: The paper contains no theorems or formal proofs. Its claims are empirical and are evaluated in Section 4.

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: [Yes]

   Justification: Section 3 specifies the models, hook placement, replacement values, task generation, validation check, metrics, split procedure, controls, and statistics. Appendix F identifies every result file. The provenance record gives exact model revisions, file hashes, package versions, seeds, and verification-run measurements. The public repository supplies the pipeline source and every result file, so a reader can reproduce the runs or recheck the reported statistics from the saved per-head and per-example values without rerunning a model.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
   - If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
   - Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
   - While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example

(a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
(b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
(c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
(d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: [Yes]

   Justification: The code and data-generation scripts are public at `https://github.com/TheClassicTechno/InterpretabilityNLPProject`. The repository contains the experiment pipelines, the statistical helpers, the saved result files for every reported run, the verification reruns, the figure script, the number-verification script, and the claim-to-evidence map, with a README giving the pinned environment and exact commands. No external dataset is needed or shipped: all task families are generated in memory from the seeds recorded in each result file by `src/dataset.py`.

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).
   - Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: [Yes]

   Justification: No model is trained. Section 3 gives the split sizes and procedure, fixed top-$k = 5$ candidate count, 2,000 bootstrap resamples, and 1,000 control draws. These settings were fixed before results were inspected; the partition and sequence-control analyses added for the camera-ready version were specified in writing before they were run. Each result file in the public repository records the full configuration used to produce it, including seeds, per-split example counts, and task parameters.

Guidelines:

- The answer [N/A] means that the paper does not include experiments.
- The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
- The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: [Yes]

   Justification: Tables 1–6 and Figure 1 report 95% percentile bootstrap intervals from 2,000 resamples or add-one-corrected Monte Carlo $p$-values from 1,000 matched-control draws. Section 3 defines both procedures. The $p = 0.078$ and $p = 0.274$ specificity results are not treated as positive findings, and the 20-partition analysis is described as a sensitivity analysis because the partitions overlap.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: [Yes]

   Justification: Section 3 reports one 10-core Apple M5 CPU, 24 GB unified memory, no GPU, and about 1.5 GB of local storage. Measured wall times and peak resident memory are 198.99 s and 1.94 GiB for the primary GPT-2 pipeline, 57.62 s and 1.90 GiB for DistilGPT2, 428.56 s and 3.22 GiB for GPT-2 at $N = 500$, and 301.43 s (1.90 GiB) and 82.08 s (1.63 GiB) for the camera-ready partition and sequence-control analyses on GPT-2 and DistilGPT2. The Qwen2.5-1.5B-Instruct pipeline used 11,976 CPU-seconds and 9.0 GiB (Appendix C). The provenance record lists every run with its command and measurements.

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.

- The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.
- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: [Yes]

   Justification: The authors reviewed the NeurIPS Code of Ethics. The experiments use public pretrained models and synthetic text. They involve no human subjects and no scraped or personal data.

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: [N/A]

    Justification: The paper tests the validity of an existing analysis technique. It does not introduce a model, deployed system, or new generative capability.

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: [N/A]

Justification: The paper releases no model weights and no scraped data. It uses small synthetic copying, addition, sequence-continuation, and IOI prompts generated at run time, and the ablations are runtime hooks, so no modified pretrained model is produced or released. The repository holds only analysis code, saved result files, and figures; the pretrained models are loaded from their existing public sources rather than redistributed.

Guidelines:

- The answer [N/A] means that the paper poses no such risks.
- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

Answer: [Yes]

Justification: The work uses GPT-2 weights released under the MIT License, DistilGPT2 and Qwen2.5-1.5B-Instruct weights released under the Apache License 2.0, and Hugging Face `transformers` v4.57.3 under the Apache License 2.0. The IOI prompts follow the published templates of Wang et al. [4], which are cited. The models and software version are named in Section 3, and their exact revisions are recorded in the provenance record. The repository ships no weights, and its code is released under the MIT License.

Guidelines:

- The answer [N/A] means that the paper does not use existing assets.
- The authors should cite the original paper that produced the code package or dataset.
- The authors should state which version of the asset is used and, if possible, include a URL.
- The name of the license (e.g., CC-BY 4.0) should be included for each asset.
- For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
- If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
- For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
- If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

Answer: [Yes]

Justification: Section 3 documents the synthetic task generators (copying, arithmetic, and ordered-sequence continuation), including their templates, sizes, seeds, and validity check, and Section 4.5 documents the IOI generator. The generator code is in `src/dataset.py` in the public repository, together with its README, the licensing terms, and the saved outputs of every reported run. The generated text contains no personal or sensitive data.

Guidelines:

- The answer [N/A] means that the paper does not release new assets.
- Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
- The paper should discuss whether and how consent was obtained from people whose asset is used.
- At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: [N/A]

    Justification: The work involves no crowdsourcing or human subjects. Section 3 describes the synthetic data generation.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
    - According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: [N/A]

    Justification: No human subjects research was conducted.

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: [Yes]

    Justification: AI coding and writing assistants helped implement the pipeline and edit the manuscript under human review. The human author defined the research questions and checked reported values against the saved results. No LLM is part of the evaluated method. This assistance is disclosed for transparency.

Guidelines:

- The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
- Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.